



**北京邮电大学**

Beijing University of Posts and Telecommunications

# **Internet of Things Engineering Practices**

## **Lab Report**

**Experiment Title**

**RFID Basic Operation**

**Group Number**

**No. 1**

**Group Members**

|    |               |            |
|----|---------------|------------|
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**Date**

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1 EXPERIMENT OBJECTIVE

- (1) Understanding of the composition, structure and function of the RFID system.
- (2) Learning to set up the SDK of RFID system and connecting the reader-transponder module to RFID software. Understanding of the working mode of RFID system.
- (3) Analyzing the data formats of EPC tag.

2 EXPERIMENT PROCEDURE

**Step 1.** Read the *XinTaiRadio\_User\_Guide* handbook and learn how to install *XinTaiRadio* software.

**Step 2.** Connect some of the RFID modules, antenna, power etc to the computer as shown in figure 1.

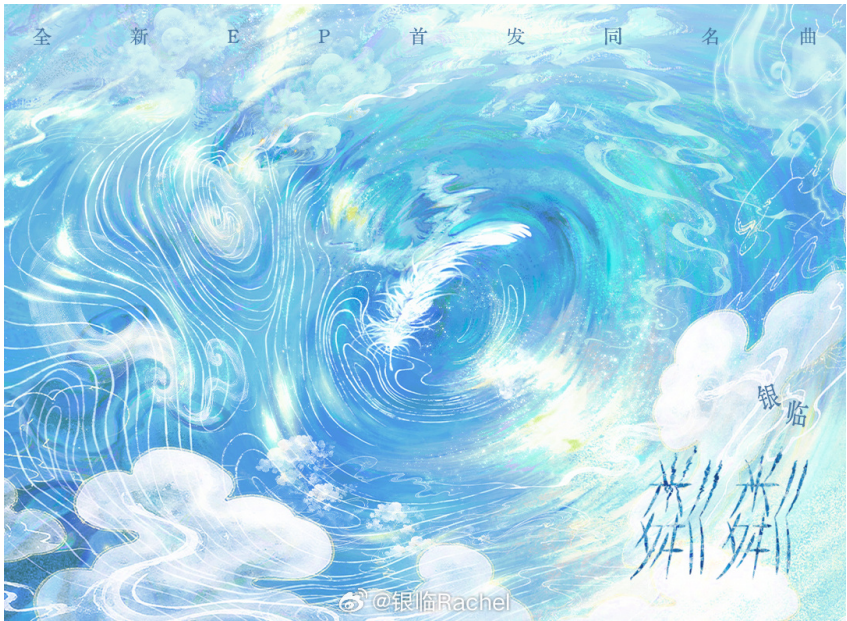


Fig. 1. Demo Picture in  $\LaTeX$

**Step 3.** When all of these components come together, use *XinTaiRadio* software to do inventory, reading and writing to the tags.

RFID technology has been widely adopted in various applications [1]. The EPC global standard provides a framework for RFID implementations [2].

3 EXPERIMENT RESULTS

In this section, describe your experimental results and observations.

3.1 RFID Tag Reading

Describe the results of reading RFID tags, including EPC codes and other relevant data. Modern RFID systems operate at different frequencies, each with specific advantages, as shown in table 1.

Table 1. RFID Tag Reading Results

| Tag ID           | Distance (cm) | Signal Strength (dBm) |
|------------------|---------------|-----------------------|
| E200001234567890 | 50            | -45                   |
| E200001234567891 | 75            | -52                   |
| E200001234567892 | 100           | -58                   |
| E200001234567893 | 125           | -65                   |
| E200001234567894 | 150           | -72                   |

3.2 Data Analysis

Analyze the format and structure of the data obtained from the RFID tags. Security considerations in RFID systems are critical [3].

4 DISCUSSION

Discuss the implications of your findings and any challenges encountered during the experiment. Recent advances in IoT have expanded RFID applications [4].

5 CONCLUSION

Summarize the key findings and what you learned from this experiment. This experiment demonstrated the basic operation of RFID systems and provided hands-on experience with RFID reader-transponder modules.

REFERENCES

[1] Roy Want. An introduction to rfid technology. *IEEE Pervasive Computing*, 5(1):25–33, 2006.

[2] EPCglobal Inc. Epc radio-frequency identity protocols class-1 generation-2 uhf rfid protocol for communications at 860 mhz - 960 mhz. Standard Specification Version 1.2.0, EPCglobal Inc., 2008.

[3] Ari Juels. Rfid security and privacy: A research survey. *IEEE Journal on Selected Areas in Communications*, 24(2):381–394, 2006.

[4] Luigi Atzori, Antonio Iera, and Giacomo Morabito. The internet of things: A survey. *Computer Networks*, 54(15):2787–2805, 2010.